SOC Lab Report: Account Compromise Detection using Splunk

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1 Abstract

This report demonstrates detection of an account compromise scenario in a SOC environment using Splunk. The attack involves multiple failed login attempts followed by a successful login, indicating that the attacker successfully guessed the credentials. The report covers simulation, detection, investigation, response, and reporting aligned with MITRE ATT&CK.

2 Objective

To detect account compromise by correlating failed and successful login events using Splunk.

3 Lab Setup

- Attacker Machine: Kali Linux
- Victim Machine: Windows 11
- SIEM Tool: Splunk (Ubuntu Server)
- Network: Internal Virtual Lab

4 MITRE ATT&CK Mapping

- Tactic: Initial Access
- Technique: Brute Force (T1110)

5 Attack Simulation

The attack was simulated by generating multiple failed login attempts followed by a successful login.

5.1 Steps Performed

- Entered incorrect password multiple times
- Logged in successfully with correct credentials

5.2 Screenshot

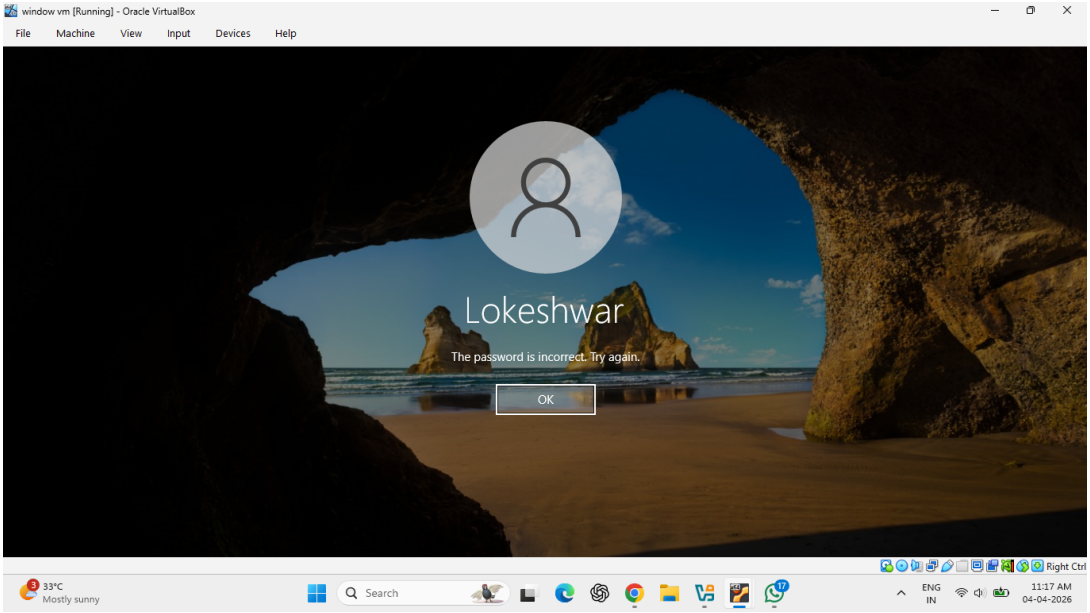

Figure 1: Multiple Failed Logins followed by Successful Login

6 Detection in Splunk

Account compromise is detected by correlating failed (4625) and successful (4624) login events.

6.1 Detection Query

```
index=windows (EventCode=4624 OR EventCode=4625)
| stats count(eval(EventCode=4625)) AS failed,
        count(eval(EventCode=4624)) AS success
        by user, Source_Network_Address
| where failed > 5 AND success > 0
```

6.2 Explanation

The query identifies users with multiple failed login attempts followed by at least one successful login from the same IP address.

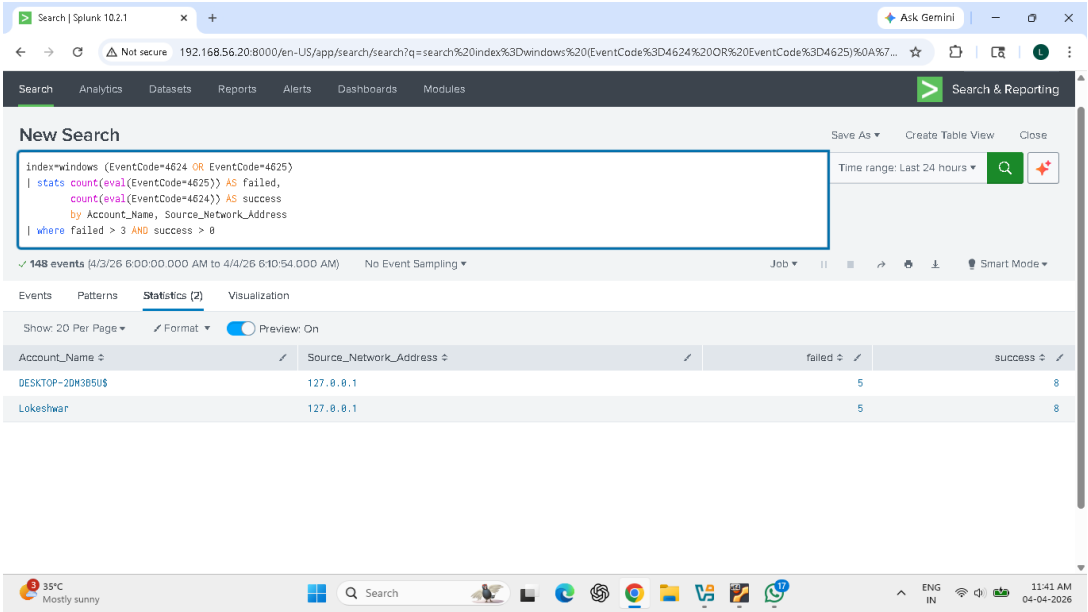

Figure 2: Detection of Account Compromise in Splunk

6.3 Screenshot

7 Investigation

Further analysis was conducted to confirm the compromise and understand the attack sequence.

7.1 Timeline Query

```
index=windows user="TARGET_USER"
| sort _time
| table _time, EventCode, Source_Network_Address
```

7.2 Findings

- Multiple failed login attempts observed
- Followed by successful login (EventCode 4624)
- Same source IP involved in both events

7.3 Screenshot

7.4 Screenshot

8 Response Actions

Immediate actions were taken to secure the compromised account.

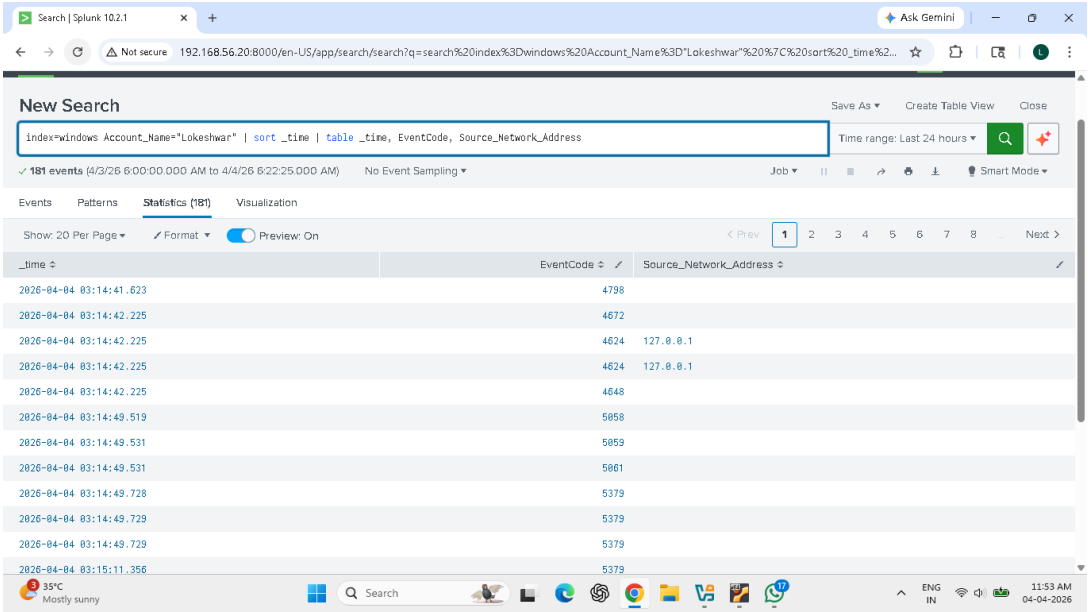

Figure 3: Timeline showing failed and successful login events

- Disabled compromised account
- Reset user password
- Blocked attacker IP

8.1 Commands Used

```
net user <username> /active:no
net user <username> NewPassword@123
New-NetFirewallRule -DisplayName "Block-Attacker" -Direction Inbound -Remo
```

8.2 Explanation

The compromised account was disabled to prevent further misuse. The password was reset and the attacker IP was blocked.

8.3 Screenshot

9 Incident Reporting

9.1 Incident Summary

An account compromise was detected where multiple failed login attempts were followed by a successful login.

9.2 Impact

Unauthorized access to the system was achieved by the attacker.

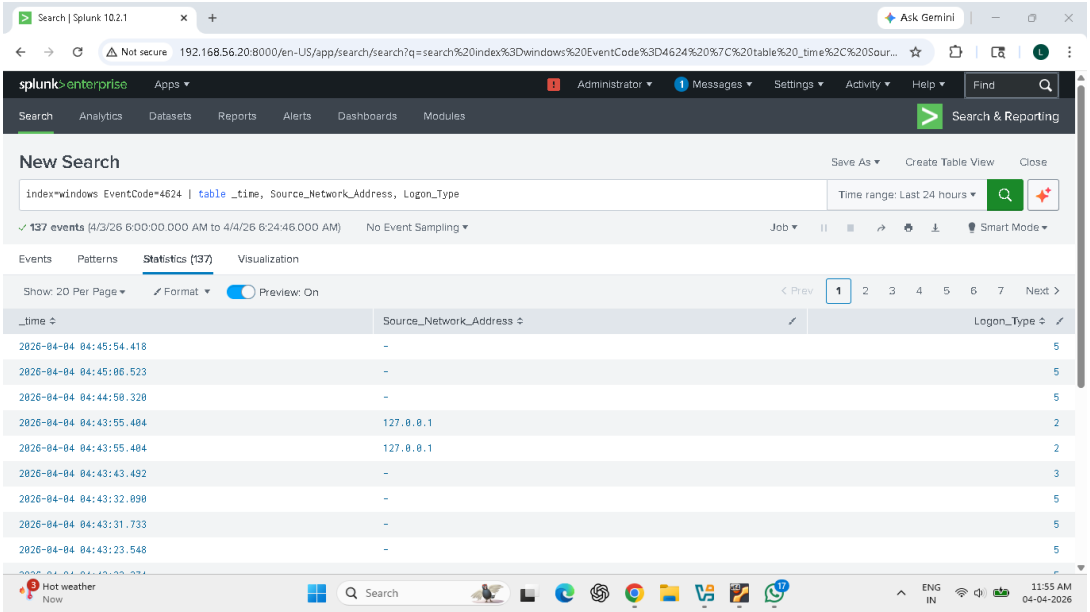

Figure 4: Timeline showing failed and successful login events

9.3 Conclusion

The compromise was successfully detected using event correlation in Splunk and mitigated through immediate response actions.

10 Conclusion

This scenario demonstrates the importance of correlating multiple events in Splunk to identify real security incidents beyond simple detection.

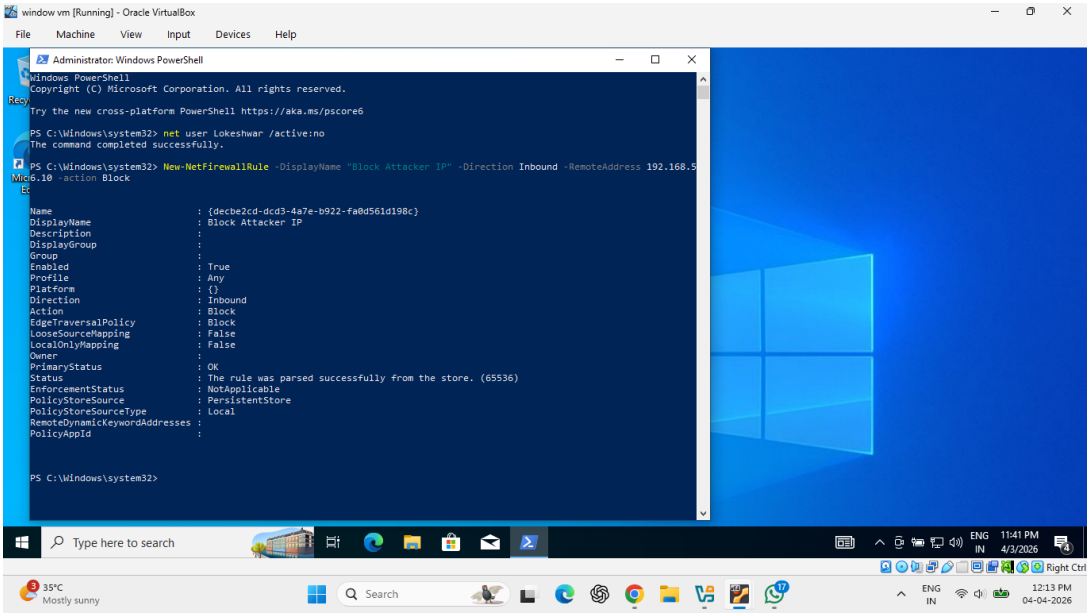

Figure 5: Response Actions - Account Disabled and IP Blocked